

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 2 – Seminar on Emerging Technologies

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 2 – Seminar on Emerging Technologies
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	N.Kalyan Reddy	Reg. No.:	192424232
Date:	22-08-2026	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Seminar on Emerging Technologies

Instructions to Students:

Students must select the topic strictly according to the Serial Number (S.No.) order assigned to them.

S.No.	Seminar Topic
1	Generative AI and Large Language Models (LLMs)
2	AI Agents and Autonomous AI Systems
3	Multimodal Artificial Intelligence
4	Explainable and Trustworthy AI
5	Edge AI and Intelligent Edge Computing
6	Federated Learning and Privacy-Preserving AI
7	Deep Learning for Computer Vision
8	AI-Powered Medical Image Analysis
9	Natural Language Processing with Transformer Models
10	Retrieval-Augmented Generation (RAG)
11	AI in Cybersecurity and Threat Detection

12	Zero Trust Architecture and Modern Cybersecurity
13	Post-Quantum Cryptography
14	Quantum Computing and Quantum Algorithms
15	Quantum Machine Learning
16	Cloud-Native Computing and Microservices
17	Serverless Computing and Function-as-a-Service
18	Edge Computing and Fog Computing
19	Cloud Security and Confidential Computing
20	Digital Twins and Intelligent Simulation
21	Internet of Things (IoT) and Smart Systems
22	Industrial Internet of Things (IIoT)
23	5G/6G Networks and Next-Generation Communication
24	Blockchain Technology and Decentralized Applications
25	Web3 and Decentralized Internet
26	Smart Contracts and Blockchain Applications
27	Augmented Reality (AR) and Virtual Reality (VR)
28	Extended Reality (XR) and Immersive Technologies
29	Robotics and Autonomous Robots
30	Human–Robot Interaction and Collaborative Robots
31	Autonomous Vehicles and Intelligent Transportation Systems
32	Smart Cities and AI-Driven Urban Technologies
33	Metaverse Technologies and Virtual Worlds
34	Brain–Computer Interfaces (BCI)
35	Wearable Computing and Smart Wearable Devices
36	Green Computing and Sustainable IT
37	AI for Software Engineering and Code Generation
38	DevSecOps and Intelligent CI/CD Pipelines
39	Digital Forensics and AI-Based Cybercrime Investigation
40	Privacy-Enhancing Technologies and Data Protection

Seminar Presentation Structure:

1. **Introduction to the Emerging Technology**
2. **Need and Motivation**
3. **Working Principle / Architecture**
4. **Key Components and Technologies**
5. **Applications / Real-World Use Cases**
6. **Advantages and Limitations**
7. **Recent Developments and Trends**
8. **Challenges and Future Scope**
9. **Case Study / Example**
10. **Conclusion and References**

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Technical Content & Subject Knowledge (25%)	Comprehensive and accurate content; demonstrates excellent understanding of the emerging technology	Good technical understanding with minor gaps	Basic understanding with limited technical depth	Incomplete content or major technical errors
Organization & Presentation Structure (15%)	Excellent logical flow with clear introduction, content, conclusion, and effective transitions	Well-structured with minor organizational issues	Adequately structured but some sections lack clarity	Poorly structured and difficult to follow
Examples, Diagrams & Case Study (20%)	Excellent use of relevant diagrams, examples, applications, and real-world case studies	Good use of relevant examples and diagrams	Limited examples/diagrams with basic explanation	Inappropriate or insufficient examples/diagrams
Communication & Presentation Skills (20%)	Very clear, confident, engaging delivery with excellent eye contact, voice modulation, and time management	Clear and confident presentation with minor issues	Understandable but lacks confidence or engagement	Unclear delivery, excessive reading, or poor time management
Question Handling & Overall Performance (20%)	Answers questions accurately and confidently; demonstrates strong overall knowledge	Answers most questions correctly	Answers basic questions with some difficulty	Unable to answer questions or demonstrate adequate understanding

Criteria	Mark
Technical Content & Subject Knowledge (25%)	/5
Organization & Presentation Structure (15%)	/5
Examples, Diagrams & Case Study (20%)	/5
Communication & Presentation Skills (20%)	/5
Question Handling & Overall Performance (20%)	/5
TOTAL	/25

1. Introduction to the Emerging Technology

Smart cities use digital technologies, connected infrastructure and intelligent analytics to improve urban services and quality of life. Intelligent parking is an important smart-city application because parking availability changes continuously and drivers may waste time searching for vacant spaces. The proposed Intelligent AI-Based Smart Parking and Vehicle Guidance System combines Internet of Things (IoT) sensors, cameras, artificial intelligence, GPS, cloud/fog computing and mobile services to detect parking availability, predict demand and guide drivers to suitable spaces.

The system is designed as an integrated smart-mobility service rather than only a parking-slot detector. It can provide real-time occupancy information, reservations, digital payments, navigation assistance and parking utilization analytics.

2. Need and Motivation

- Drivers spend time searching for vacant parking spaces, especially during peak periods.
- Circling vehicles can increase congestion, fuel consumption and unnecessary emissions.
- Traditional parking systems may not provide accurate real-time availability.
- Parking demand varies by time, day, events, weather and location, creating a need for prediction.
- Administrators need centralized occupancy information and utilization reports for better planning.
- A connected system can combine detection, prediction, guidance, reservation and payment into one service.

3. Working Principle / Architecture

The system begins by collecting occupancy information from IoT sensors and cameras installed in parking areas. Edge or gateway devices preprocess the data and forward relevant information to the cloud or fog platform. AI models analyze current and historical occupancy to estimate demand. A guidance engine then identifies suitable vacant spaces using availability and location information. The mobile application can display the recommended space, support reservation and initiate digital payment. Updated occupancy data is continuously fed back into the platform so that the system remains synchronized.

Proposed Intelligent AI-Based Smart Parking Architecture

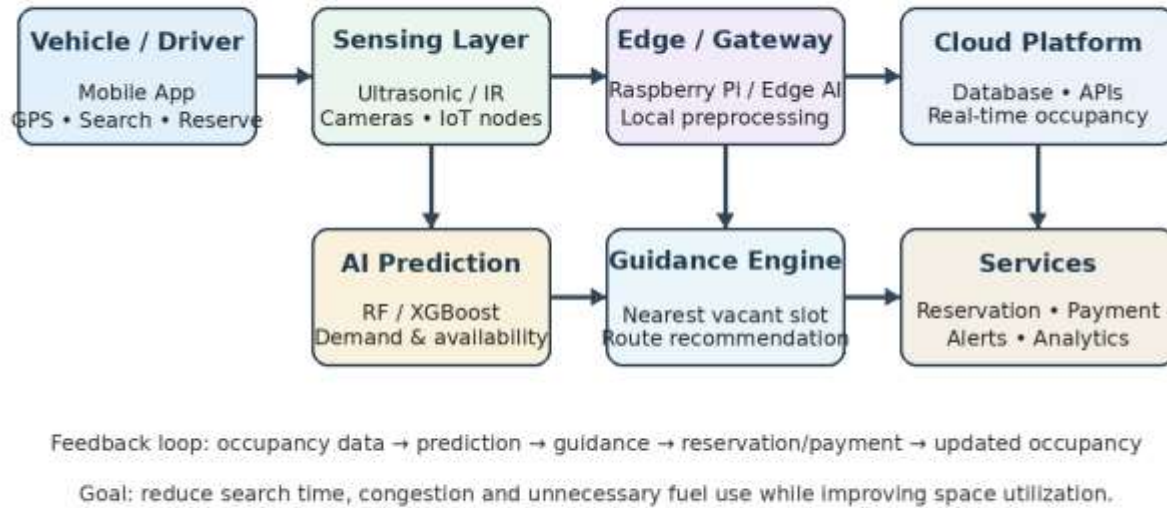


Figure 1. Proposed architecture of the Intelligent AI-Based Smart Parking and Vehicle Guidance System.

4. Key Components and Technologies

Component	Role
IoT Sensors	Ultrasonic, IR or other occupancy sensors detect whether individual spaces are occupied.
Computer Vision	Cameras and vision models can identify vehicles and classify parking-slot occupancy.
Artificial Intelligence	Machine-learning models predict parking demand and support intelligent allocation.
Edge/Fog Computing	Local processing reduces latency and unnecessary transmission of raw data.
Cloud Platform	Stores historical information, provides APIs, analytics and centralized administration.
GPS and Maps	Supports navigation and route guidance to recommended parking spaces.
Mobile Application	Allows users to search, reserve, receive guidance

	and monitor parking status.
Digital Payment	Supports secure online parking payments.
Admin Dashboard	Provides occupancy monitoring, utilization reports and operational analytics.

5. Applications / Real-World Use Cases

- Smart-city public parking and municipal parking zones.
- Shopping malls and commercial complexes.
- University and college campuses.
- Hospitals and healthcare campuses.
- Office and IT parks.
- Airports, railway stations and transport hubs.
- Event venues where parking demand changes sharply during peak periods.
- Residential communities with controlled parking.

6. Advantages and Limitations

Advantages

- Reduces the time required to search for parking.
- Provides real-time parking availability.
- Supports demand prediction and better space utilization.
- Improves driver convenience through guidance and reservation.
- Can reduce unnecessary vehicle circulation and associated fuel use.
- Provides administrators with occupancy and utilization analytics.

Limitations

- Sensors and cameras require installation, maintenance and calibration.
- Computer-vision performance can be affected by lighting, weather and occlusion.
- Network failures can affect real-time services if no edge fallback is provided.
- AI models require representative historical data and periodic updating.
- Camera-based systems introduce privacy and data-governance considerations.
- Large-scale deployment can involve significant infrastructure and integration costs.

7. Recent Developments and Trends

Recent smart-parking research increasingly combines IoT with fog/edge computing, machine learning and computer vision. A 2025 Scientific Reports study evaluated a real-time IoT and fog-computing parking system using a practical case study at SDU University Park and reported benefits in responsiveness, network usage and space utilization compared with cloud-based processing.

[cite](#) [turn0search0](#)

Recent work is also moving toward hybrid AI-powered parking systems that dynamically allocate spaces using real-time demand, user preferences and contextual information. Research published in 2025 describes AI/IoT approaches for hospitals, colleges, offices and shopping malls, with emphasis on demand prediction and personalized recommendations.

[cite](#) [turn0search2](#)

Vision-based systems are another important direction. A 2025 Engineering Applications of Artificial Intelligence study proposed an IoT-enabled vision approach for real-time parking-slot detection and occupancy classification, aiming to reduce dependence on costly physical sensors.

[cite](#) [turn0search3](#)

Another 2025 study combined IoT, Raspberry Pi, camera-based detection and mobile guidance, showing how smart parking can integrate real-time detection with user navigation and access control.

[cite](#) [turn0search7](#)

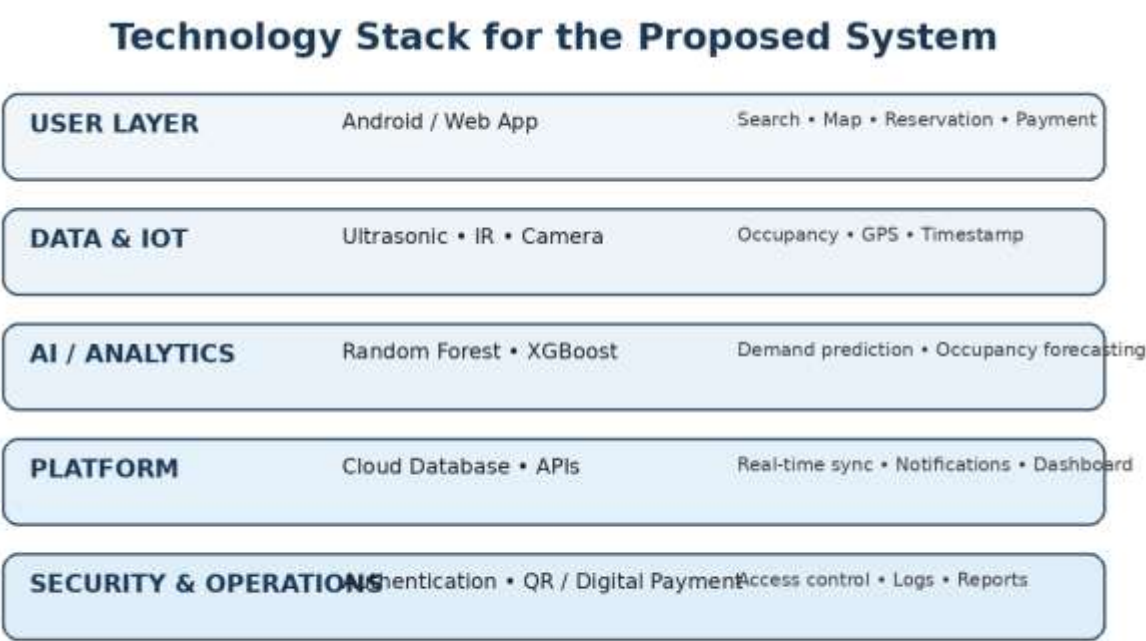


Figure 2. Technology stack for the proposed smart parking system.

8. Challenges and Future Scope

1. Improve robustness of occupancy detection across different lighting, weather and parking layouts.
2. Develop adaptive AI models that learn from changing parking patterns and special events.
3. Use edge AI for low-latency decisions while retaining cloud analytics for long-term planning.
4. Integrate secure digital identity, payment and access-control mechanisms.
5. Apply privacy-preserving methods to camera and vehicle data.
6. Integrate electric-vehicle charging availability with parking guidance.
7. Connect parking data with wider intelligent transportation and smart-city platforms.
8. Evaluate systems using search-time reduction, prediction error, latency, utilization, energy use and user satisfaction.

9. Case Study / Example

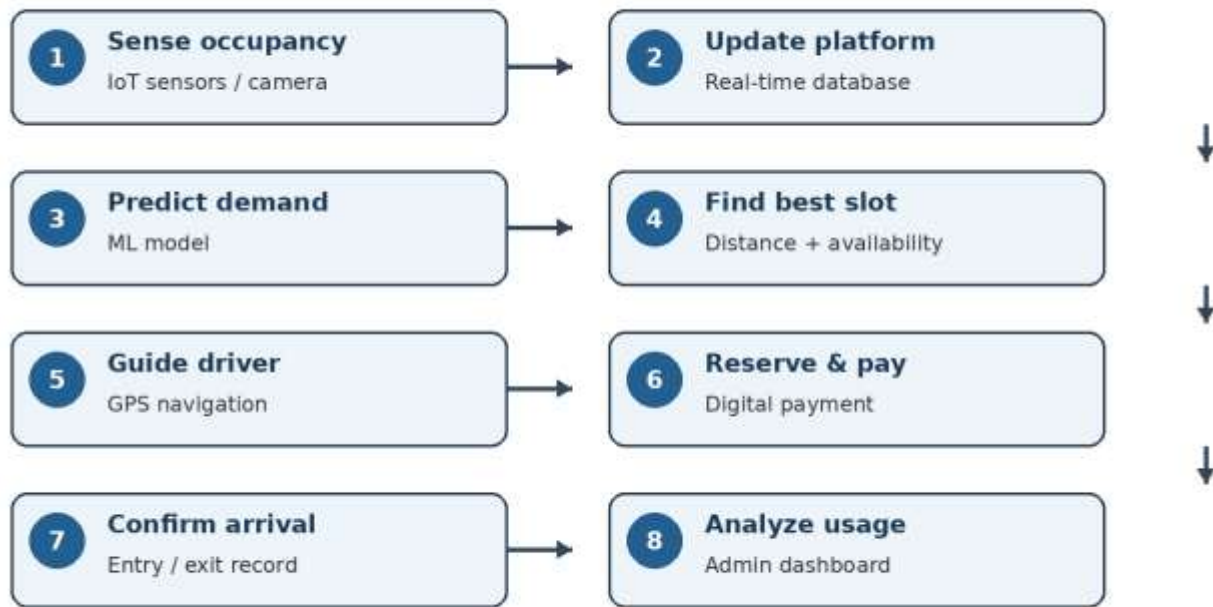
Case Study: SDU University Park Fog-Based Smart Parking

A 2025 practical study developed and evaluated an IoT-based smart parking system using a hierarchical fog architecture. Parking sensors and edge/fog processing were used to provide real-time information and reduce the need to send all data to a central cloud. The study evaluated latency, execution cost, execution time, network usage and energy consumption, and included a real-world deployment at SDU University Park. The reported results showed the fog-based approach to be more efficient and responsive than the cloud-based alternative, particularly for peak-hour parking management.

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This case study supports the proposed system because it demonstrates the value of local processing and real-time availability information. The proposed project extends this idea by combining prediction, driver guidance, reservation, payment and administrator analytics in a single workflow.

Smart Parking and Vehicle Guidance Workflow



The workflow combines real-time sensing, AI prediction, navigation, reservation and analytics.

Figure 3. End-to-end workflow from occupancy sensing to guidance, reservation, payment and analytics.

10. Conclusion and References

The Intelligent AI-Based Smart Parking and Vehicle Guidance System represents an application of smart-city and AI-driven urban technology. By combining IoT sensing, computer vision, AI prediction, edge/fog computing, GPS, cloud services and mobile applications, the system can improve parking visibility and driver decision-making. The major future opportunity is to develop a reliable, privacy-aware and scalable platform that works across different parking environments and integrates parking with broader intelligent transportation services.

[1] M. A. Ala'anzy, A. Abilakim, R. Zhanuzak, and L. Li, "Real time smart parking system based on IoT and fog computing evaluated through a practical case study," *Scientific Reports*, vol. 15, 33483, 2025. DOI: 10.1038/s41598-025-15507-6.

[2] "AI-Powered Hybrid Smart Parking: Optimizing Parking Management Across Diverse Applications in Smart Cities," *Procedia Computer Science*, vol. 258, pp. 1524–1535, 2025. DOI: 10.1016/j.procs.2025.04.385.

- [3] H. Alkhudhayr, “Internet of things based parking slot detection and occupancy classification for smart city traffic management,” *Engineering Applications of Artificial Intelligence*, vol. 152, 110802, 2025. DOI: 10.1016/j.engappai.2025.110802.
- [4] “Enhanced IoT-Enabled Smart Parking system with RFID Integration and Line Assistance,” *Transportation Research Procedia*, vol. 84, pp. 346–353, 2025. DOI: 10.1016/j.trpro.2025.03.082.
- [5] M. Alymani, L. A. Almoqhem, D. A. Alabdulwahab, A. A. Alghamdi, H. Alshahrani, and K. Raza, “Enabling smart parking for smart cities using Internet of Things (IoT) and machine learning,” *PeerJ Computer Science*, vol. 11, e2544, 2025. DOI: 10.7717/peerj-cs.2544.
- [6] S. Hudda et al., “A WSN and vision based smart, energy efficient, scalable, and reliable parking surveillance system with optical verification at edge for resource constrained IoT devices,” *Internet of Things*, vol. 28, 101346, 2024. DOI: 10.1016/j.iot.2024.101346.
- [7] S. Deepan Kumar et al., “IoT-Enabled Cloud-Integrated Smart Parking System with Real-Time Monitoring and AI-Based Space Optimization for Next-Gen Mobility,” *SAE Technical Paper 2026-28-0113*, 2026. DOI: 10.4271/2026-28-0113.

Prepared according to the Seminar Presentation Structure in the provided CSA10 CO5 Assessment Tool